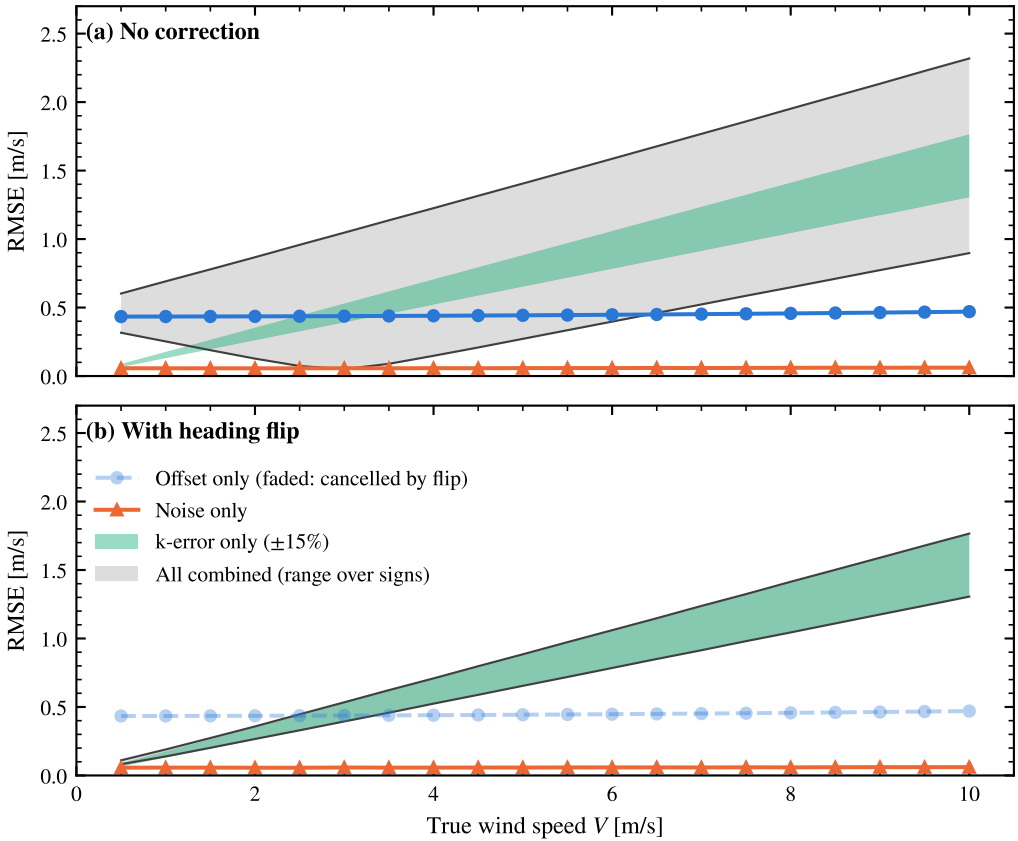

Hattenberger (measured): m = 0.554 kg, k = 0.153 N·s/m. Tilt offset b = ±0.7°, noise σ = 0.5° per sample, n_eff = 30 (placeholders, D4). Estimator k off by ±15% (D7, from the 14-20 % wind-tunnel vs flight gap in Hattenberger et al. 2023). 10,000 trials per wind speed, seed 0.